

Tempus Stylus What's New

Product Version 26.10

May 2026

© 2017-2026 Cadence Design Systems, Inc. All rights reserved.
Printed in the United States of America.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Trademarks: Trademarks and service marks of Cadence Design Systems, Inc. (Cadence) contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence's trademarks, contact the corporate legal department at the address shown above or call 1-800-862-4522.

All other trademarks are the property of their respective holders.

Restricted Print Permission: This publication is protected by copyright law and international treaties and contains trade secrets and proprietary information owned by Cadence. Unauthorized reproduction or distribution of this publication, or any portion of it, may result in civil and criminal penalties. Except as expressly permitted below, this publication may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way, including automated processes, training or services involving a large language model, foundation model, deep machine learning, generative artificial intelligence, or any other similar process or technology, without prior written permission from Cadence. Unless otherwise agreed to by Cadence in writing, customers are granted permission to print one (1) hard copy of this publication subject to the following conditions:

1. The publication may be used solely for personal, informational, and noncommercial purposes;
2. The publication may not be modified in any way;
3. Any copy of the publication or portion thereof must include all original copyright, trademark, and other proprietary notices and this permission statement; and
4. Cadence reserves the right to revoke this authorization at any time, and any such use shall be discontinued immediately upon written notice from Cadence.

Disclaimer: Information in this publication is subject to change without notice and does not represent a commitment on the part of Cadence. Except as may be explicitly set forth in such agreement, Cadence does not make, and expressly disclaims, any representations or warranties as to the completeness, accuracy or usefulness of the information contained in this document. Cadence does not warrant that use of such information will not infringe any third party rights, nor does Cadence assume any liability for damages or costs of any kind that may result from use of such information. Cadence is committed to using respectful language in our code and communications. We are also active in the removal and/or replacement of inappropriate language from existing content. This product documentation may however contain material that is no longer considered appropriate but still reflects long-standing industry terminology. Such content will be addressed at a time when the related software can be updated without end-user impact.

Restricted Rights: Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

Contents

About This Manual	7
How This Document Is Organized	7
Related Documents	7
1	8
Stylus Release Overview	8
Release 26.10 Overview	8
Release Highlights	8
Timing Analysis Enhancements	8
GBA Pessimism Reduction Improvements	9
Hierarchical Flow Enhancements	9
Runtime and Memory Improvements	10
Accuracy Improvements	10
Infrastructure Enhancements	10
New Text Commands	11
New Command Parameters	12
New Root Attributes	13
Obsolete Commands	15
Removed from Software	15
2	16
Default Behavior Changes	16
Timing Analysis	17
Timing Report Format Default Change in PBA Mode	17
3	18
Timing Modeling	18
Release 26.10 Enhancements	19
4	20
Delay Calculation	20
Release 26.10 Enhancements	21
Support for Derate-Based Advanced Multi-Input Switching (AMIS-Derate)	21
Ability to Set Advanced Variation Settings	22
Ability to Scale Via Resistance using Scaling Factor	22

Added New Wire Variation Attributes	22
5	23
Timing Analysis and Reporting	23
Timing Enhancements	24
Release 26.10 Enhancements	25
Timing Analysis Enhancements	25
Support for Hierarchical Timing Context Analysis	25
Added New Attributes	26
Added New timing_spatial_derate_distance_mode Option	26
Timing Constraint Enhancements	26
set_path_adjust_group Command Enhancements	27
Added New create_generated_clock Parameter	27
timing_generated_clocks_inherit_ideal_latency Attribute Enhanced	27
Added New Parameters	28
Ability to Query Clock Pins and Clock Source Latency	28
IPD Analysis Enhancements	29
Enhanced write_ipd_scope Command	29
Scope Analysis Enhancements	30
create_boundary_model Command Enhancements	30
Signal Integrity Enhancements	31
Release 26.10 Enhancements	31
Added merge_noise_reports Command	31
Added New Command Parameters	31
Added New si Category Attributes	31
Report Enhancements	33
Release 26.10 Enhancements	33
Added New Report Attributes	33
report_constraint Command Enhancements	33
report_timing Collection Report Output Enhancements	34
Added New report_case_analysis Parameters	34
6	35
Distributed Static Timing Analysis	35
Release 26.10 Enhancements	36
Supported Parameters in DSTA	36
7	37

Manual ECO	37
Release 26.10 Enhancements	38
Added New create_net Parameter	38
Added New delete_nets Parameter	38
8	39
Signoff ECO	39
Release 26.10 Enhancements	40
Added New Attributes to opt_signoff Category	40
9	41
Other Enhancements	41
Release 26.10 Enhancements	41
flow Attribute Category Enhancement - New Attribute for Identifying Active Plugin Step	42
New Command for Reporting Connections Between Two Hinsts	42
Enhancement to Restrict Saving the Database in the ECO Batch Mode	42
flowtool Attribute Category Enhanced	42
check_ac_limit Attribute Category Enhanced	42
Added New Attribute	43
route_special Attribute Category Enhanced	44
New Commands to Abstract Cloned Instances of Large Design Modules	44
New Command to Zoom into Highlighted Timing Paths	44
Clock Tree Debugger Enhancements	44
Support for Coloring Based on Maximum Fanout Constraint	44
Support for Displaying Partition Hierarchy	45
Signal Edge Coloring Enhanced	46
Support for Displaying Skew Group Source Pin	47
Support for Zooming into a Selected Item in the Clock Tree Viewer	47
Support for Displaying Skew Group Source Pin	47
Quick Access to Object Attributes	47
Color Map Changes for Setup and Hold Slack	48

About This Manual

This manual provides information about Cadence® Tempus™ Timing Solution.

How This Document Is Organized

This manual is organized into chapters that cover broad areas of Tempus functionality. Each chapter contains topics that address one or more of the following areas:

- New functionality in Tempus and enhancements made to existing forms and commands to support a new feature.
- Changes in default behavior, name changes to existing commands and forms, and syntax changes.
- Major documentation changes, such as a new chapter or substantial reorganization.

Related Documents

For more information about Tempus, see the following documents. You can access these and other Cadence documents with the Cadence Help online documentation system.

- [*Tempus Known Problems and Solutions*](#)
Describes important Cadence Change Requests (CCRs), and provides workarounds for known problems.
- [*Tempus Text Command Reference*](#)
Describes the Tempus text commands, including syntax and examples.
- [*Tempus Menu Reference*](#)
Describes the Tempus graphical user interface.
- [*Tempus User Guide*](#)
Provides information about using various Tempus features.
- [*Tempus Foundation Flows User Guide*](#)
Describes the Cadence-recommended procedures for performing basic STA and SI signoff, MMMC signoff, and CPF-based MMMC signoff using Cadence® Tempus™ Timing Solution.
- README file
Contains installation, compatibility, and other prerequisite information, including a list of Cadence change requests (CCRs) that were resolved in this release. You can read this file online at downloads.cadence.com.

Stylus Release Overview

- [Release 26.10 Overview](#)
 - [Release Highlights](#)
 - [Timing Analysis Enhancements](#)
 - [GBA Pessimism Reduction Improvements](#)
 - [Hierarchical Flow Enhancements](#)
 - [Runtime and Memory Improvements](#)
 - [Accuracy Improvements](#)
 - [Infrastructure Enhancements](#)
 - [New Text Commands](#)
 - [New Command Parameters](#)
 - [New Root Attributes](#)
 - [Obsolete Commands](#)
 - [Removed from Software](#)

Release 26.10 Overview

Release Highlights

Timing Analysis Enhancements

Derate-Based Advanced Multi-Input Switching (AMIS-Derate) Analysis

The new analysis feature, Derate-based AMIS (AMIS-Derate), is now available in this release. A simulation-based advanced multiple-input switching (AMIS-Simulation) analysis, already available in Tempus previous releases, can model MIS effects on both delay and slew. As a result of modeling MIS effects on slew, enabling AMIS-simulation can be very pessimistic. Due to the detailed simulation performed for each instance, the runtime overhead can be high.

To mitigate these shortcomings, the AMIS-derate feature has been introduced. AMIS-derate pre-

characterizes MIS impact on multi-input cells as delay derates for different slew/load combinations and uses these derates in timing analysis. AMIS-derate applies MIS impact only to the base delay component. AMIS-derate maintains accuracy similar to AMIS-simulation and provides better runtime, debuggability, and reporting capabilities.

For more information, refer to "[Support for Derate-Based Advanced Multi-Input Switching \(AMIS-Derate\) Analysis](#)".

Virtual IDX (VIDX) SPEF Support in Tempus 3DIC Analysis

Tempus STA now supports reading VIDX SPEF generated by Quantus. The VIDX SPEF models the capacitance impact between two dies.

GBA Pessimism Reduction Improvements

Arc Bounding Box Mode for Spatial Derates

Spatial derate calculation uses the `chip_size` mode in GBA, which is conservative. The new arc bounding box mode has been introduced in Tempus STA, which reduces GBA spatial derate pessimism by limiting bounding box calculations to all timing paths through an instance. The results in reduced GBA-PBA gap and reduced GBA pessimism result in improved PBA runtime.

Improved GBA Waveform Compression

The enhanced GBA waveform compression reduces pessimism in graph-based analysis by refining the transition waveform representation to better reflect true signal behavior. This preserves waveform accuracy while minimizing conservatism. And also results in a reduced GBA-PBA gap.

Hierarchical Flow Enhancements

Interface Timing Context Generation in DSTA

Tempus now supports timing context generation in DSTA, enabling interface path reporting in hierarchical context analysis. Previously in DSTA, it was limited only to register-to-register path reporting.

Pushdown SDC Support in DSTA

Tempus supports Pushdown SDC generation in DSTA and enables both register-to-register and interface path reporting in hierarchical context analysis.

Boundary Model Generation without Additional Timer Update

Tempus has been enhanced to support boundary models generation without triggering a timer update internally. Previously, boundary models generation implicitly triggered a timer update, resulting in performance overhead. This enhancement is primarily beneficial for hierarchical flows that require both timing context and boundary models generation in the same run.

Runtime and Memory Improvements

Faster `update_timing` with Selective in-Memory SI Reporting

This enhancement improves the `update_timing` command performance by optimizing single-CPU runtime and multi-CPU scalability with the `si_delay_enable_report true` setting.

New Advanced SI Analysis

The new advanced SI analysis changes how attacker slews are computed by establishing a signoff accurate base delay calculation before SI analysis begins. This replaces the earlier approach that relied on a fast pre-SI analysis attacker slew computation. Beyond the accuracy benefits, it also offers runtime benefits from enhanced reuse provided by signoff-accurate base iteration.

Accuracy Improvements

Consolidated Delay Calculation Framework

This enhancement in delay calculation aligns key simulation steps within a single, consistent computation framework across timing and SI analysis. By synchronizing these steps, the analysis achieves improved accuracy, greater robustness, and better correlation with SPICE results.

Infrastructure Enhancements

Python Shell Support

In this release, Python Shell is supported in Tempus STA. This can be used for data analysis, visualization, and algorithm development. Python shell can be accessed in two ways:

- **Dedicated Python shell mode:** In this mode, the Tempus console is changed into a Python interpreter where Python code can be specified directly. All the Tempus Python commands are available as Python functions.
- **Embedded Python execution (within Tcl):** In this mode, blocks of Python code can be inserted within Tcl scripts using the `eval_python {}` command. This mode can be used to leverage Python's data processing and scripting capabilities for specific tasks in a Tcl-based flow.

Parallel `report_timing` (PRT) Framework

Tempus STA now supports a robust and flexible framework that allows parallel execution of multiple `report_timing` commands. This helps to reduce overall reporting runtime when the right set of `report_timing` commands is specified for PRT execution.

New Text Commands

The following table lists the commands that were added to the Tempus software in this release. The second column identifies the corresponding chapter of the *Tempus Stylus Text Command Reference* where the command is documented.

New Commands	Chapter
<code>db_hinst_connection</code>	Basic Database Access Tcl Commands and Attributes
<code>get_selectively_expanded_modules</code>	Basic Database Access Tcl Commands and Attributes
<code>merge_noise_reports</code>	Signal Integrity Commands and Attributes
<code>set_advanced_variation_mode</code>	Signal Integrity Commands
<code>set_selectively_expanded_modules</code>	Basic Database Access Tcl Commands and Attributes
<code>zoom_highlighted_timing_path</code>	Timing Debug Commands

New Command Parameters

The following table lists the parameters that were added to the Tempus Timing Solution software. The second column identifies the chapter of the Tempus Text Command Reference where the command is documented.

New Parameters	Chapter
<code>create_boundary_model</code> <code>-skip_disabled</code> <code>-include_userpins</code> <code>-latch_effort_low</code>	Timing Analysis Commands and Attributes
<code>create_generated_clock</code> <code>-preinvert</code>	Timing Constraint Commands
<code>create_net</code> <code>-module</code>	Manual ECO Commands and Attributes
<code>delete_nets</code> <code>-module</code>	Manual ECO Commands and Attributes
<code>report_case_analysis</code> <code>-include_function</code> <code>-include_ref_name</code>	Timing Analysis Commands and Attributes
<code>report_constraint</code> <code>-drv_abs_margin</code> <code>-drv_ratio_margin</code>	Timing Analysis Commands and Attributes
<code>report_noise</code> <code>-merge</code>	Signal Integrity Commands and Attributes
<code>report_noise_propagation</code> <code>-style</code>	Signal Integrity Commands and Attributes

<code>reset_timing_derate</code> <code>-transition</code>	Timing Analysis Commands and Attributes
<code>set_path_adjust_group</code> <code>-fall</code> <code>-fall_from</code> <code>-fall_through</code> <code>-fall_to</code> <code>-rise</code> <code>-rise_from</code> <code>-rise_through</code> <code>-rise_to</code>	Timing Analysis Commands and Attributes
<code>set_sense</code> <code>-clock_leaf</code>	Timing Constraint Commands
<code>write_ipd_scope</code> <code>-include_derates</code> <code>-include_constraints</code>	Timing Analysis Commands and Attributes
<code>write_spice_deck</code> <code>-scale_mode</code>	Signal Integrity Commands and Attributes

New Root Attributes

The following table lists the root attributes that were added to the Tempus Timing Solution software. The second column identifies the category in which the attribute has been added and the third column identifies the corresponding chapter of the *Tempus Stylus Text Command Reference* where the root attribute is documented.

New Root Attributes	Category	Chapter
check_ac_limit_additional_individual_violation_report check_ac_limit_check_thermal_aware_em check_ac_limit_em_cdf_percentage check_ac_limit_ircx_models check_ac_limit_lifetime_rating_factor_file check_ac_limit_report_unannotated_shapes check_ac_limit_simulation_net check_ac_limit_simulation_net_file check_ac_limit_top_scope_ignore_block_internal_nets_on_boundary_path	check_ac_limit	Check Comm and Attribu
delaycal_via_corner_sigma_multiplier delaycal_via_correlation_type delaycal_via_reset delaycal_separate_params_for_via delaycal_wire_wire_cell_variation_correlation delaycal_wire_use_nsigma_coupling_capacitance	delaycal	Signa Integri Comm and Attribu
flow_plugin_canonical_current	flow	Flowk Comm and Attribu
flowtool_schedule_flow_immediate	flowtool	Flowk Comm and Attribu
opt_signoff_high_effort_cell_list opt_signoff_smart_mmmc_force_keep_hold_view_list opt_signoff_smart_mmmc_force_keep_setup_view_list	opt_signoff	opt_si Categ Attribu
route_special_pad_pin_split_from_center	route_special	Power Route Categ Attribu
si_virtual_inter_die_attacker_slew	si	Signa Integri Comm and Attribu

<code>timing_analysis_enable_inter_power_domain</code> <code>timing_consider_late_arrival_for_ssi_derate</code> <code>timing_disable_get_clocks_for_current_instance</code> <code>timing_enable_hierarchical_get_nets_support</code> <code>timing_extract_model_enable_output_pin_attributes</code> <code>timing_extract_model_half_cycle_non_interface_paths</code> <code>timing_extract_model_synchronous_clocks_as_skew_checks</code> <code>timing_extract_model_write_function_for_combinational_arcs</code> <code>timing_ipd_xdomain_path_filtering_threshold</code> <code>timing_library_ignore_combinational_arc_mismatch_tables</code> <code>timing_report_enable_qpin_as_start_for_begin_end_pair</code> <code>timing_report_enable_write_twf_third_party_for_input_pins</code> <code>timing_report_limit_max_path_count</code> <code>timing_report_skip_icg_from_all_fanin_fanout_collection</code>	<code>timing</code>	Timing Analy: Comm and Attribu
<code>write_db_restrict_under_eco_batch_mode</code>	<code>write_db</code>	Import Expor Comm and Attribu

Obsolete Commands

Removed from Software

The following command has been removed from the software:

- `reset_sdc`

Default Behavior Changes

- [Timing Analysis](#)
 - [Timing Report Format Default Change in PBA Mode](#)

Release 26.10 Enhancements

The following default changes have been made.

Timing Analysis

Timing Report Format Default Change in PBA Mode

Previous Behavior	New Behavior
Earlier, by default timing reports (<code>report_timing</code> command) did not display PBA column headers.	In this release, by default, the timing reports will display PBA column headers for nsigma, mean, sigma, and standard deviation details.
	Impact: This default change might impact parsing of reports in some cases.

Timing Modeling

- [Release 26.10 Enhancements](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added New Attributes

The following new attributes have been added:

- `timing_extract_model_half_cycle_non_interface_paths`: When set to `true`, the software models half-cycle non-interface paths as MPW (minimum pulse width) checks in ETM. By default, this global variable is set to `false`.
- `timing_extract_model_enable_output_pin_attributes`: Specifies the attributes (e.g., `always_on`, `is_isolated`) to enable on output pins during ETM generation. You can use this global variable when backward compatibility with older ETM behavior is required. By default, this global variable is set to `none`.
- `timing_extract_model_synchronous_clocks_as_skew_checks`: When set to `true`, the software additionally captures constraints in ETM for reg-to-reg paths, which are launched and captured by different synchronous clocks within the block. The software converts the slack of such paths into a skew-rising/falling constraint requirement between the two clock pins at the IP boundary. By default, this global variable is set to `false`.
- `timing_extract_model_write_function_for_combinational_arcs`: When set to `true`, the software writes the function attribute on ports with combinational arcs. By default, this global variable is set to `false`.

Refer to [timing category attributes](#).

Delay Calculation

- [Release 26.10 Enhancements](#)
 - [Support for Derate-Based Advanced Multi-Input Switching \(AMIS-Derate\)](#)
 - [Ability to Set Advanced Variation Settings](#)
 - [Ability to Scale Via Resistance using Scaling Factor](#)
 - [Added New Wire Variation Attributes](#)

Release 26.10 Enhancements

The following enhancements have been made.

Support for Derate-Based Advanced Multi-Input Switching (AMIS-Derate)

A simulation-based advanced multiple-input switching (AMIS-Simulation) analysis can model MIS effects on both delay and slew. As a result of modeling MIS effects on slew, enabling AMIS-simulation can be very pessimistic. Due to the detailed simulation performed for each instance, the runtime overhead can be high.

To mitigate these shortcomings, the AMIS-derate feature has been introduced. AMIS-derate pre-characterizes MIS impact on multi-input cells as delay derates for different slew/load combinations and uses these derates in timing analysis. AMIS-derate applies MIS impact only to the base delay component. AMIS-derate maintains accuracy similar to AMIS-simulation and provides better runtime, debuggability, and reporting capabilities.

- To enable derate-based MIS, you can use the following setting:

- `set_db timing_analysis_multi_input_switching_mode 3`

When MIS-derate is enabled, Tempus will automatically disable AMIS-simulation.

- To generate MIS-derate reports (in PBA mode only), the `report_timing -format` parameter supports the `retime_mis_derate` option.
- The `get_property` (and `report_property`) command supports the following new properties:

Property	Added to Object
<code>has_mis_fall</code>	• <code>timing_arc</code> • <code>lib_timing_arc</code>
<code>has_mis_rise</code>	• <code>timing_arc</code> • <code>lib_timing_arc</code>
<code>mis_alignments_factor_min_rise</code>	<code>lib_cell</code>
<code>mis_alignments_factor_min_fall</code>	<code>lib_cell</code>

For more information on AMIS-Derate, refer to "[Derate-Based Advanced Multi-Input Switching \(AMIS-Derate\) Analysis](#)" section in the *Tempus Stylus User Guide*.

Ability to Set Advanced Variation Settings

The Tempus software allows you to set up advanced variation settings. The following new command has been added:

- `set_advanced_variation_mode`: Sets the advanced variation scope to configure generic effect specific settings.

Ability to Scale Via Resistance using Scaling Factor

The Tempus software allows you to scale via resistance from SPEF that is based on the user-specified scaling factor. The following new attributes have been added:

- `delaycal_via_corner_sigma_multiplier`: Provides the scaling multiplier for corner scaling. This attribute allows you to specify per via sigma multiplier and handles the advanced via variation scaling. *Default:* " "
- `delaycal_via_correlation_type`: Specifies inter-layer correlation values. *Default:* `corner`
- `delaycal_via_reset {true | false}`: Resets via variation modes to their default value. *Default:* `false`
- `delaycal_separate_params_for_via {true | false}`: Allows to create separate parameters for via variation. *Default:* `false`

For more information, refer to [delaycal Category Attributes](#).

Added New Wire Variation Attributes

The following wire variation attributes have been added:

- `delaycal_wire_wire_cell_variation_correlation`: Controls whether to RSS or add the random cell variation and wire variation for arrivals and slack. *Default:* `false`
- `delaycal_wire_use_nsigma_coupling_capacitance`: Allows to use the mean+nSigma coupling capacitance. *Default:* `false`

For more information, refer to [delaycal Category Attributes](#).

Timing Analysis and Reporting

- [Timing Enhancements](#)
- [Signal Integrity Enhancements](#)
- [Report Enhancements](#)

Timing Enhancements

- Release 26.10 Enhancements
 - Timing Analysis Enhancements
 - Support for Hierarchical Timing Context Analysis
 - Added New Attributes
 - Added New timing_spatial_derate_distance_mode Option
 - Timing Constraint Enhancements
 - set_path_adjust_group Command Enhancements
 - Added New create_generated_clock Parameter
 - timing_generated_clocks_inherit_ideal_latency Attribute Enhanced
 - Added New Parameters
 - Ability to Query Clock Pins and Clock Source Latency
 - IPD Analysis Enhancements
 - Enhanced write_ipd_scope Command
 - Scope Analysis Enhancements
 - create_boundary_model Command Enhancements

Release 26.10 Enhancements

The following enhancements have been made.

Timing Analysis Enhancements

Support for Hierarchical Timing Context Analysis

The Tempus hierarchical timing context feature provides top-down analysis techniques to model accurate block boundary constraints from the top hierarchy or top STA environment. Tempus provides a bottom-up (Boundary Models) and top-down (Context Models) hierarchical timing analysis/closure solution to address the challenges encountered during timing and glitch analysis/closure of large hierarchical designs. The context models for lower-level modules are generated from top level analysis to perform accurate block level timing and glitch analysis. The context data is generated in a proprietary format and can be read only in Tempus.

To use this feature, the following setting is required:

```
timing_analysis_type ocv
```

The context analysis feature is not supported in `best_case_worst_case` timing analysis mode.

To support this feature, the following features have been added:

New Commands

- `set_timing_context_clock_mapping`
- `set_timing_context_view_mapping`
- `report_timing_context_clock_mapping`
- `compare_timing_context`

New Attributes

- `timing_hierarchical_context_continue_on_top_block_clock_mismatch`: Allows loading of the timing context model with unmapped clocks. *Default:* `false`
- `timing_enable_hier_context_unmapped_clock_analysis`: When set to `true`, for unmapped block clocks, the software enables timing analysis based on the block SDC constraints set on block ports. *Default:* `false`
- `timing_context_port_based_clock_mapping`: When set to `true`, clock mapping is done depending solely on clock property of the top and block clock for each clock and data port separately. *Default:* `false`
- `timing_context_apply_port_sdc_exceptions`: When set to `true`, the software will consider path

exceptions on block constraints in the hierarchical context flow. *Default:* `false`

For more information, refer to [timing category attributes](#).

For more information, refer to "[Hierarchical Timing Context Analysis](#)" section in the *Tempus Stylus User Guide*.

Added New Attributes

The following new attributes have been added:

- `timing_consider_late_arrival_for_ssi_derate`: When set to `true`, the software uses the late arrival time for window overlapping evaluation during simultaneous switching input (SSI) analysis. *Default:* `false`
- `timing_enable_hierarchical_get_nets_support`: When set to `true`, the software enables the hierarchical segment support for the `get_nets` command. *Default:* `false`
- `timing_disable_get_clocks_for_current_instance`: When set to `true`, the software disables `current_instance` command support on the `get_clocks` command. *Default:* `false`
- `timing_ipd_xdomain_path_filtering_threshold`: When this attribute is used, the specified threshold (in ns) will be used in the path filtering criterion during the native IPD xdomain solution. *Default:* `0.003`

For more information, refer to [timing category attributes](#).

Added New `timing_spatial_derate_distance_mode` Option

By default, spatial derate calculation uses the `chip_size` mode in GBA, which is conservative. The new arc bounding box mode has been introduced in Tempus STA, which reduces GBA spatial derate pessimism by limiting bounding box calculations to all timing paths through an instance. The results in reduced GBA-PBA gap and reduced GBA pessimism result in improved PBA runtime.

The following new option has been added to the `timing_spatial_derate_distance_mode` attribute:

- `arc_bounding_box`

You can use this option when

the `timing_enable_pessimistic_cpnr_for_reconvergent_clock_paths` attribute is set to `true`.

For more information, refer to [timing category attributes](#).

Timing Constraint Enhancements

set_path_adjust_group Command Enhancements

The `set_path_adjust_group` command supports the following edge-specific parameters:

- `-fall`: Applies `set_path_adjust_group` at the falling edge of the signal on the path endpoints.
- `-fall_from <from_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the from pins.
- `-fall_through <through_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the through pins.
- `-fall_to <to_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the to pins.
- `-rise`: Applies `set_path_adjust_group` at the rising edge of the signal on the path endpoints.
- `-rise_from <from_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the from pins.
- `-rise_through <through_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the through pins.
- `-rise_to <to_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the to pins.

Added New create_generated_clock Parameter

The following `create_generated_clock` parameter has been added:

- `-preinvert`: Inverts the polarity of a master clock at the specified source pin.

timing_generated_clocks_inherit_ideal_latency Attribute Enhanced

Earlier, when the `set_clock_latency` command was used to define network latency for ideal clock mode at intermediate pins of a clock network, Tempus ignored the latency. In this release, the `timing_generated_clocks_inherit_ideal_latency` attribute has been enhanced to accept Enum values that allow including the closest applied network latency for ideal inheritance. The following new options have been added:

- `actual_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and propagated delays for the path from this point to the assertion point of the generated clock.
- `ideal_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and ideal delays for the path from this point to the assertion point of the generated clock.

Note: When the `-source` parameter is used to specify the latency of a generated clock, the software

completely overrides any inherited latency.

By default, this attribute is set to `false`.

Added New Parameters

The following parameters have been added:

- `set_sense -clock_leaf`: When specified, the software propagates clock phases to the specified pin.
- `reset_timing_derate -transition`: Resets the transition derate values.

Ability to Query Clock Pins and Clock Source Latency

The `get_property` (and `report_property`) command has been enhanced to query on clock source objects and latency. The following new properties have been added:

Property	Object Type	Description
<code>source_latency_pins</code>	clock	Returns a collection of pins and ports in the source latency network of a generated clock. This property is undefined for master clocks. You can index by clock name using the <code>get_property -clock</code> command parameter. <i>Return type:</i> collection
<code>is_clock_source_latency</code>	pin port net	Returns a value of <code>true</code> if a pin/port or any of the leaf pins of a net is part of a clock source latency network. <i>Return Type:</i> Boolean

<code>is_ref_clock_pin</code>	<code>pin</code> <code>lib_pin</code>	Returns a value of <code>true</code> for: - Valid instance pin object and on an active clock pin where a clock signal reaches, and sequential cell was not disabled by disabled timing arcs or case analysis. Any pin with a clock check or trigger arc will also be reported if an active clock phase is present on that pin. - Clock pin on an inferred clock-gating cell. - Valid <code>lib_pin</code> object, if there is at least one instance of this <code>lib_pin</code> in the design, and if the <code>is_clock_pin</code> property is set to <code>true</code>. - Returns a value of <code>false</code> if clock propagation is disabled with the <code>set_disable_timing</code> or <code>set_sense - stop_propagation</code> command. <i>Return Type:</i> Boolean
<code>is_clock_source_pin</code>	<code>pin</code> <code>port</code>	Returns a value of <code>true</code> for a valid instance pin or port object, if a clock is declared with that clock source at the pin/port. <i>Return Type:</i> Boolean
<code>is_clock_pin</code>	<code>lib_pin</code>	Returns a value of <code>true</code> if at least one of the timing arcs for setup/hold exists. Also, if equivalent timing constraints are present in the library definition of the library pin. <i>Return Type:</i> Boolean

IPD Analysis Enhancements

Enhanced `write_ipd_scope` Command

Tempus allows identification of inter-power-domain (IPD) paths with more than two power-domain crossings. The `write_ipd_scope` command has been enhanced to support multiple power-domain paths during early IPD checks. The following new options have been added to the `-early_structural_checks` parameter:

- `multiple_pd_data_paths`
- `multiple_pd_clock_paths`
- `multiple_pd_paths`

The following `write_ipd_scope` command parameters have been added:

- `-include_constraints`: Allows to write IPD scope with victim constraints.
- `-include_derates`: Allows to write IPD scope with timing derates.

For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus Stylus User Guide*.

Scope Analysis Enhancements

`create_boundary_model` Command Enhancements

The following `create_boundary_model` command parameters have been added:

- `-skip_disabled`: Skips tracing of logic through disabled arcs (user or constant disabled) in the boundary model.
- `-include_userpins`: Includes the specified pins in the boundary model. No fanin/fanout tracing is performed for the specified pins.
- `-latch_effort_low`: Skips tracing of logic through latch aggressors that are not part of the interface logic. The aggressors in the fanout of latch aggressors will use an infinite timing window. By default, the entire fanin cone of latch aggressors is preserved to ensure accurate latency adjustment, which may increase the size of the boundary model.

The following parameters have been removed:

- `-absolute_tolerance`
- `-exclude_constraints`
- `-include_full_netlist`
- `-validate_local_cpu`

Signal Integrity Enhancements

- [Release 26.10 Enhancements](#)
 - [Added merge_noise_reports Command](#)
 - [Added New Command Parameters](#)
 - [Added New si Category Attributes](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added merge_noise_reports Command

The Tempus software allows you to generate merged noise reports for the specified views. The following new command has been added:

- `merge_noise_reports`

Added New Command Parameters

The following new parameters have been added:

- `report_noise_propagation -style {default | extended}`: Provides an enhanced report format that shows new fields - UDN and User annotated. You can specify the following options:
 - `default`: Prints the ROP column and VDD (in mV) in the output report.
 - `extended`: Reports the drivers with annotated glitches as propagation sources.
- `report_noise -merge`: Generates a merged report that shows only the worst glitch slack for each pin level combination across views.
- `write_spice_deck -scale_mode string`: Allows you to specify the UMC_URI scale mode option.
Default: effective

Added New si Category Attributes

The following attributes have been added:

- `si_virtual_inter_die_attacker_slew value`: Specifies the virtual inter-die attacker slew value.
Default: 2.5e-11

- `si_enable_advanced_flow {true | false}`: Enables advanced SI analysis. *Default:* `false`

For more details, refer to the [si category attributes](#).

Report Enhancements

- [Release 26.10 Enhancements](#)
 - [Added New Report Attributes](#)
 - [report_constraint Command Enhancements](#)
 - [report_timing Collection Report Output Enhancements](#)
 - [Added New report_case_analysis Parameters](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added New Report Attributes

The following report attributes have been added:

- `timing_report_skip_icg_from_all_fanin_fanout_collection`: When set to `true`, the software skips ICG clock pins from the `all_fanin` (or `all_fanout`) command collection when queried with startpoints/endpoints. By default, this global variable is set to `false`.
- `timing_report_enable_qpin_as_start_for_begin_end_pair`: Enables reporting of multiple paths with different q-pins having the same clock pin. By default, this global variable is set to `false`.
- `timing_report_enable_write_twf_third_party_for_input_pins`: When set to `true`, the software writes timing information for input pins in third-party TWF file. By default, this global variable is set to `false`.
- `timing_report_limit_max_path_count`: Sets the maximum number of violations (of data/clock checks) to be reported. By default, this global variable is set to `2000000` (integer).

For more details, refer to the [timing Category Attributes](#).

report_constraint Command Enhancements

The following `report_constraint` command enhancements have been made:

- Added new parameters:
 - `-drv_abs_margin`: Specifies the absolute DRV margin value applied to obtain the threshold

transition.

- `-drv_ratio_margin`: Specifies the ratio DRV margin value applied to obtain the threshold transition.

- Added new `-drv_fields` option:

`hnet`: Reports hierarchical nets. When the `timing_enable_hierarchical_get_nets_support` attribute is set to `true`, the software enables support for displaying hierarchical net names. For more details, refer to the [timing Category Attributes](#).

report_timing Collection Report Output Enhancements

The timing report Retime Analysis header has been enhanced for collection reports. When the `report_timing -path_type exhaustive -collection` command is used, the following tags will be printed in the report output:

- `PBA COLL`: Shows a collection of paths for which retime analysis is performed while creating the collection.
- `GBA COLL`: Shows a collection of paths for which retime analysis is not performed.

When the `report_timing -collection` parameter is specified, the GBA and PBA slack values are not shown separately by default.

Added New report_case_analysis Parameters

The following `report_case_analysis` parameters have been added:

- `-include_function`: Displays the pin function of a pin. This must be specified with the `-propagated` parameter.
- `-include_ref_name`: Displays the reference name of an instance in the report.

Distributed Static Timing Analysis

- [Release 26.10 Enhancements](#)
 - [Supported Parameters in DSTA](#)

Release 26.10 Enhancements

The following enhancements have been made.

Supported Parameters in DSTA

The following parameters are now supported in DSTA:

- `set_license_check -checkout`
- `set_license_check -wait_time`

Manual ECO

- [Release 26.10 Enhancements](#)
 - [Added New create_net Parameter](#)
 - [Added New delete_nets Parameter](#)

Release 26.10 Enhancements

The following enhancements have been made:

Added New `create_net` Parameter

The following new `create_net` parameter has been added:

- `-module`: Adds the new net to the specified Verilog module.

Added New `delete_nets` Parameter

The following new `delete_nets` parameter has been added:

- `-module`: Deletes the net from the specified Verilog module.

Signoff ECO

- [Release 26.10 Enhancements](#)
 - [Added New Attributes to opt_signoff Category](#)

Release 26.10 Enhancements

The following enhancements have been made:

Added New Attributes to `opt_signoff` Category

The following new `opt_signoff` category attributes have been added:

- `opt_signoff_smart_mmmc_force_keep_hold_view_list`: List of hold views, which have to be retained after SmartMMMC view compaction. These views represent the minimum hold views retained by the system, on top of which additional views can be retained.
- `opt_signoff_smart_mmmc_force_keep_setup_view_list`: List of setup views, which have to be retained after SmartMMMC view compaction. These views represent the minimum setup views retained by the system, on top of which additional views can be retained.
- `opt_signoff_high_effort_cell_list`: Enables the controlled utilization of high-leakage or high-power cells during signoff optimization. These cells are typically classified as `dont_use` or `dont_touch` due to their power characteristics; however, they may be beneficial in specific scenarios. When this attribute is specified, ensure that the `don't_use` constraint is set to `false` for cells included in `opt_signoff_high_effort_cell_list`.
 - In setup timing optimization, the cells are considered only if they are essential to meet timing requirements without causing a significant increase in power consumption.
 - In power optimization, the cells may be inserted if their usage leads to an overall decrease in power usage, such as enhancements in fanin/fanout cone.

This attribute allows treating the specified cells as high-effort resources, using them only when necessary to improve timing or power efficiency, while maintaining the broader design objective of reducing power impact.

Other Enhancements

- Release 26.10 Enhancements
 - flow Attribute Category Enhancement - New Attribute for Identifying Active Plugin Step
 - New Command for Reporting Connections Between Two Hinsts
 - Enhancement to Restrict Saving the Database in the ECO Batch Mode
 - flowtool Attribute Category Enhanced
 - check_ac_limit Attribute Category Enhanced
 - Added New Attribute
 - route_special Attribute Category Enhanced
 - New Commands to Abstract Cloned Instances of Large Design Modules
 - New Command to Zoom into Highlighted Timing Paths
 - Clock Tree Debugger Enhancements
 - Support for Coloring Based on Maximum Fanout Constraint
 - Support for Displaying Partition Hierarchy
 - Signal Edge Coloring Enhanced
 - Support for Displaying Skew Group Source Pin
 - Support for Zooming into a Selected Item in the Clock Tree Viewer
 - Support for Displaying Skew Group Source Pin
 - Quick Access to Object Attributes
 - Color Map Changes for Setup and Hold Slack

Release 26.10 Enhancements

The following enhancements have been made:

flow Attribute Category Enhancement - New Attribute for Identifying Active Plugin Step

In this release, a new attribute, `flow_plugin_canonical_current`, has been added to the `flow` attribute category. Use this attribute to identify the canonical plugin step currently being run, enabling accurate detection of the plugin within the Flowtool-based flows. When a plugin is not run, this attribute returns an empty value.

New Command for Reporting Connections Between Two Hinsts

In this release, a new text command, `db_hinst_connection`, has been introduced. You can use this command for reporting connections between two hinsts. The following are the parameters of this command:

```
[ -help ]  
hInst hInst
```

Enhancement to Restrict Saving the Database in the ECO Batch Mode

A new attribute, `write_db_restrict_under_eco_batch_mode`, has been introduced. You can use this attribute to prevent the database from being saved in the ECO batch mode.

flowtool Attribute Category Enhanced

In this release, a new attribute, `flowtool_schedule_flow_immediate`, has been added to the `flowtool` attribute category. When set to `true`, flowtool will start the scheduled flows immediately. When set to `false`, it waits for the software to exit before starting the scheduled flows.

check_ac_limit Attribute Category Enhanced

In this release, the following new attributes have been added to the `check_ac_limit` attribute category:

- `check_ac_limit_additional_individual_violation_report`: Specifies to generate separate violation reports for each current waveform calculation method (`rms`, `peak`, or `avg`). If this attribute is not specified, a single violation report file is generated using the `check_ac_limit_method {RMS AVG peak}` attribute.
- `check_ac_limit_check_thermal_aware_em`: Specifies to use the thermal map for EM checks. The thermal map file is specified using the `set_rail_analysis_config -read_thermal_map` parameter. The Rail Analysis engine uses the temperature specified in the thermal map file for EM analysis. EM limit has a strong temperature dependency. For a chip that has substantial temperature

variation across the chip, the Rail Analysis engine can use the temperature specified in the thermal map file to perform accurate EM analysis.

- `check_ac_limit_em_cdf_percentage`: Calculates the Cumulative Distribution Factor (CDF) derating factor based on the EM rules specified in the ICT EM file, given by the `check_ac_limit_ict_em_models` attribute. This attribute is used to calculate the failure rate and reliability of a design. The value of the CDF percentage should be between 0 and 100.
- `check_ac_limit_ircx_models`: Specifies an encrypted or unencrypted RC/EM technology (.ircx) file. The foundry-specific IRCX file, which contains the process information for extraction and EM modeling, can be used when the ICT-EM file is not available for electromigration analysis.
- `check_ac_limit_lifetime_rating_factor_file`: Specifies a lifetime rating factor file that contains `jmax_life` values for each layer. These values are defined based on temperature and lifetime pairs. When you provide this file, EM analysis will automatically select the appropriate `jmax_life` factor for each layer based on the specific temperature-lifetime combinations defined in the file.
- `check_ac_limit_report_unannotated_shapes`: Provides visibility into partial net coverage within SPEF files. When this attribute is set to `true`, the `<report_name>.rpt.unannotated_shapes` is generated. This report identifies and lists nets that contain shapes without resistance annotations or unanalyzed/partially analyzed segments. The feature helps to detect incomplete parasitic extraction coverage and improve accuracy during validation and signoff workflows. The reporting of partial net coverage works only when `check_ac_limit_effort_level` is set to `high`.
- `check_ac_limit_simulation_net`: Forces the specified nets to use the simulation mode.
- `check_ac_limit_simulation_net_file`: Forces the nets in the specified file to use the simulation mode.
- `check_ac_limit_top_scope_ignore_block_internal_nets_on_boundary_path`: Ignores the internal nets from the first instance to the first flop on the boundary path for the signal EM checking.

Added New Attribute

The following new attribute has been added:

- `timing_library_ignore_combinational_arc_mismatch_tables`: When set to `true`, the software will give preference to `timing_type` over arc tables. This will change the inferred arcs based on the `timing_type` (liberty format) or timing tables settings. A warning message will be displayed when the preference (`timing_type` or arc tables) is changed.

By default, this attribute is set to `false`.

route_special Attribute Category Enhanced

In this release, a new attribute, `route_special_pad_pin_split_from_center`, has been added to the `route_special` attribute category. Use this attribute to split the width and spacing specified by the `route_special_pad_pin_split` attribute to enable connections to the center of wide pad pins. This parameter must be used with the `route_special_pad_pin_split` attribute.

New Commands to Abstract Cloned Instances of Large Design Modules

In this release, the following new commands have been introduced:

- `set_selectively_expanded_modules`: Abstracts cloned instances of modules with a high number of instantiations, and flatten is one or more of them. This reduces memory usage and runtime significantly during the STA and ECO processes. It helps handle large designs on smaller machines and facilitates ECO implementation without the need to manually partition the design into smaller segments.
- `get_selectively_expanded_modules`: Displays the values set by the `set_selectively_expanded_modules` command.

New Command to Zoom into Highlighted Timing Paths

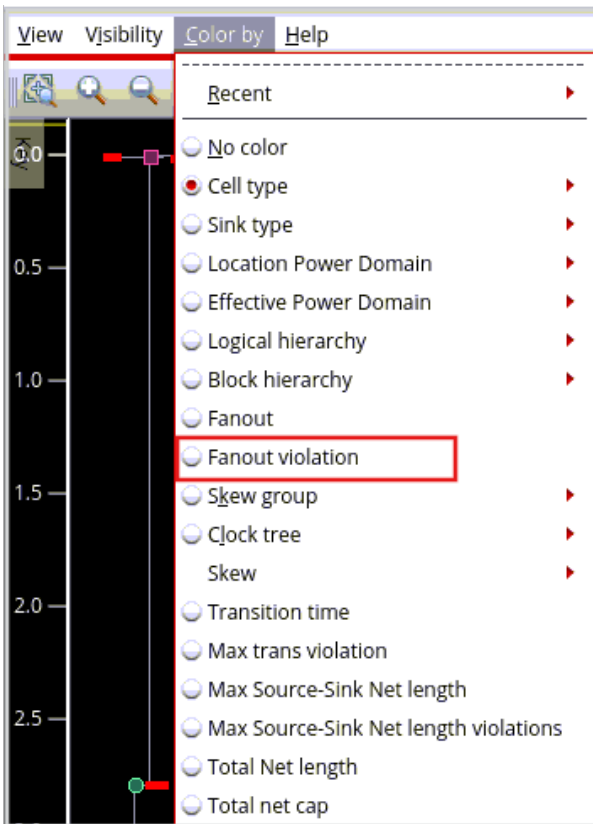
In this release, a new command, `zoom_highlighted_timing_path`, has been introduced. Use this command to zoom into highlighted timing paths in the Timing Debug form in the Layout view, and to enable sequential traversal when multiple paths are highlighted. This command reduces the number of steps required to view and navigate timing paths in the Layout view. The command is very useful for zooming into highlighted timing paths in large designs. The following are the parameters of this command:

```
[-help]  
[-previous | -next]
```

Clock Tree Debugger Enhancements

Support for Coloring Based on Maximum Fanout Constraint

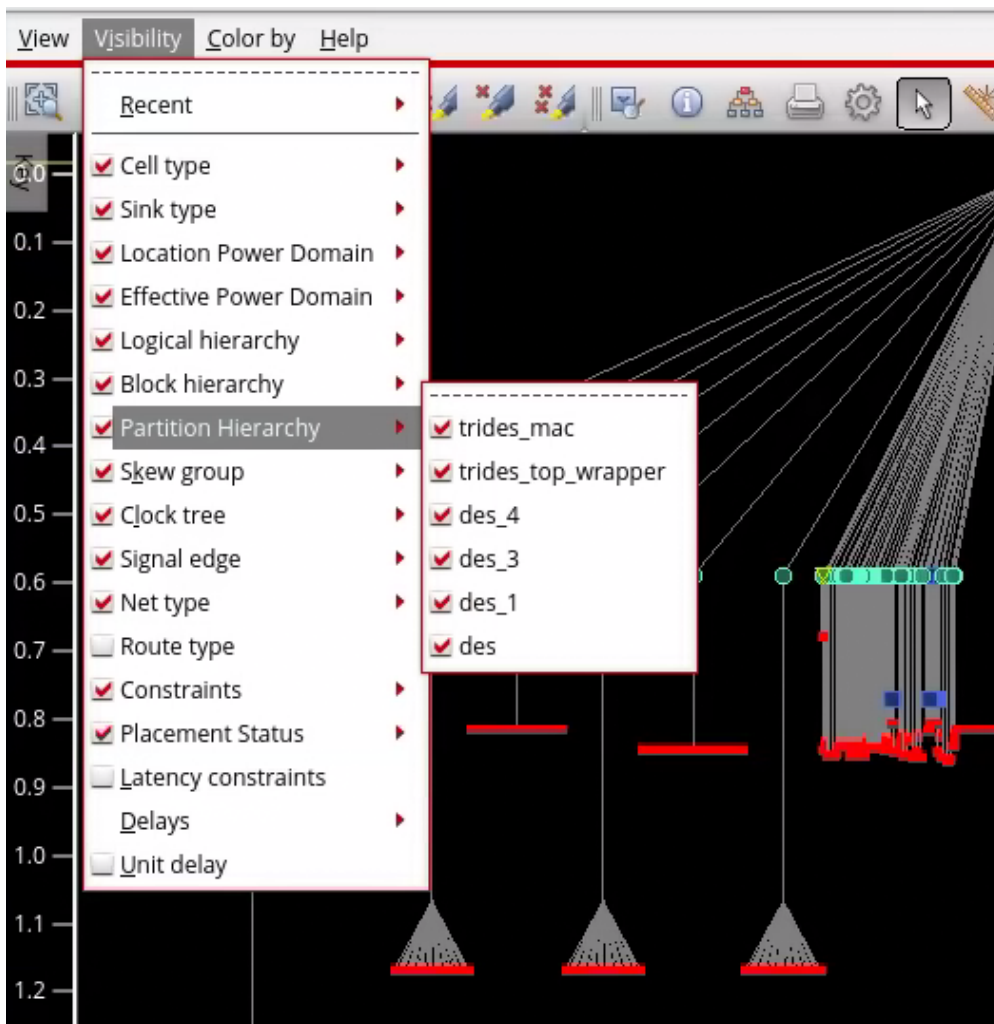
The CTD is enhanced to support coloring selected nodes based on the maximum fanout constraint. For this feature, a new option, *Fanout violation*, is added to the *Color by* menu and the *Control* panel. The tooltip shows the violation amount and target value.



For details, see the [Clock Menu](#) chapter in *Tempus Menu Reference*.

Support for Displaying Partition Hierarchy

A new option called *Partition Hierarchy* has been provided in the *Visibility* menu, *Color by* menu, and *Control* panel. This option provides a color category for each partition in the design. Each marker is drawn with the color associated with that partition.



This option is only shown in these menus for designs with partitions configured and is available in unit delay and timing-based modes.

Signal Edge Coloring Enhanced

The *Signal edge* option in the *Visibility* menu, the *Color by* menu, and the *Control* panel provide options to view and color signal edges for cells and nets. In this release, the signal edge feature is enhanced in the following ways:

- The cell coloring was previously only based on the edges found on the input and output nets. New cell coloring is based on the more accurate timing arcs.
- The *Cell* and *Net* items of the *Signal edge* option now have no color associated with them in the tree view. Only the options available within each subgroup have color options.
- The default coloring is changed to make them more colorblind friendly.
- The *Root* option is removed from the *Cell* subgroup. This option was used for both non-generated

and generated clock roots. It is replaced by a new *Unknown* option that includes non-generated root cells. The generated root cells are now colored according to the *Simple/Inverting/Non-unate* options.

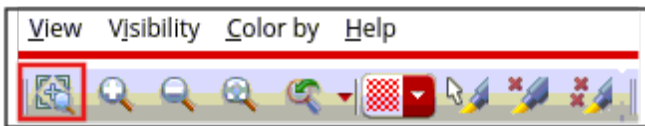
- The *Unknown* option is also added in the *Net* subgroup. It is used for nets with no skew groups.
- The *Net* subgroup coloring is enhanced to include nets connected to generator path nodes.

Support for Displaying Skew Group Source Pin

When displaying a single skew group, the source pin is typically not shown because CTD markers represent input pins, while skew groups usually start at output pins. Starting from this release, CTD is enhanced so that when coloring or displaying by skew group, a marker is treated as part of the skew group if either its input pin or the output pin of the corresponding clock node belongs to that skew group. This ensures that it remains visible when other skew groups are hidden.

Support for Zooming into a Selected Item in the Clock Tree Viewer

In this release, a new toolbar widget, *Zoom Selected*, is added to the row of widgets below the CTD menu bar. This widget allows you to zoom into a selected item in the Clock Tree Viewer without opening the *Set Preference* form. The equivalent bindkey is `Ctrl+Z`.



As part of this enhancement, the *Enable Zoom selected* option in the *Set Preference* form is renamed to *Enable Zoom layout selected*. The equivalent bindkey provided is `Ctrl+E`.

Support for Displaying Skew Group Source Pin

When displaying a single skew group, the source pin is typically not shown because CTD markers represent input pins, while skew groups usually start at output pins. Starting from this release, CTD is enhanced so that when coloring or displaying by skew group, a marker is treated as part of the skew group if either its input pin or the output pin of the corresponding clock node belongs to that skew group. This ensures that it remains visible when other skew groups are hidden.

Quick Access to Object Attributes

You can now view object attributes directly from the Clock Tree Viewer using the `⌘` bindkey. Select any object and press `⌘` to open the *Object Attributes* form, providing faster access to detailed object information without navigating menus.

Color Map Changes for Setup and Hold Slack

The color map represents the slack range from negative to positive values. In this release, it is enhanced so that white indicates the most negative slack, transitioning to red at zero slack. Positive slack is shown starting in green and progressing to purple, which represents the highest positive slack.

